

Can It Be Sorted?

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

Given an array a of n elements, determine whether it can be sorted using **at most one swap operation**.

In one swap operation, you may choose any two positions and swap their values.

The final array is considered **sorted** if it is either **non-decreasing** or **non-increasing** order.

Input

The first line contains an integer n ($1 \leq n \leq 10^5$).

The second line contains n integers a_i ($1 \leq a_i \leq 10^9$).

Output

Print YES if the array can be made sorted using at most one swap operation. Otherwise, print NO.

Examples

standard input	standard output
5 1 2 5 3 4	NO
5 1 2 3 5 4	YES